

1 - Introduction

I became interested in stock price simulation via my studies in differential equations. Having developed an interest in practical applications of mathematics, I found finance to be an ideal area to apply abstract and analytical reasoning to the real world. Moreover, my previous computer project in ODEs led me to consider stochastic differential equations.

When I learnt about Geometric Brownian Motion and its application to finance, I decided to code this project in order to explore the relationship and hone my programming capabilities. The aim is to investigate how different factors affect the outcome of the whole simulation, for example volatility and number of individual simulations, as well as explore the limitations and potential impracticalities of Geometric Brownian Motion as a model for stock price paths.

2 - The GBM Model

The Geometric Brownian Motion model of a stock price path is governed by the stochastic equation $dS_t = \mu S_t dt + \sigma S_t dW_t$ where:

- t is time,
- S_t is the price of the stock (in this case, AAPL) at time t units (in this case, trading days),
- μ is the drift or expected rate of return,
- σ is the volatility or uncertainty in how the price will evolve with time,
- $dW_t = Z\sqrt{dt}$ where Z is a standard normal random variable.

When the model is discretised for practical use, dt becomes a discrete time step. I chose dt to be equal to one trading day.

Itô's calculus yields an analytic solution which, when $dt = 1$, can be expressed recursively as $S_{t+1} = S_t \exp(\mu - \frac{1}{2}\sigma^2 + \sigma Z)$. In other words, we can simulate tomorrow's price using today's price by simulating the outcome of Z .

3 - Methodology

I used the yfinance module to fetch real historical data for the price of AAPL stock.

I chose to use data from 1 Jan 2023 to 1 Jan 2026, as this felt like a considerable length of time without going so far back as to include now-irrelevant data.

Because an aim of the project is to investigate how volatility and number of simulations affect the results of the whole simulation, these parameters can be changed by the user. (However, there is an option to estimate the volatility from historical data.)

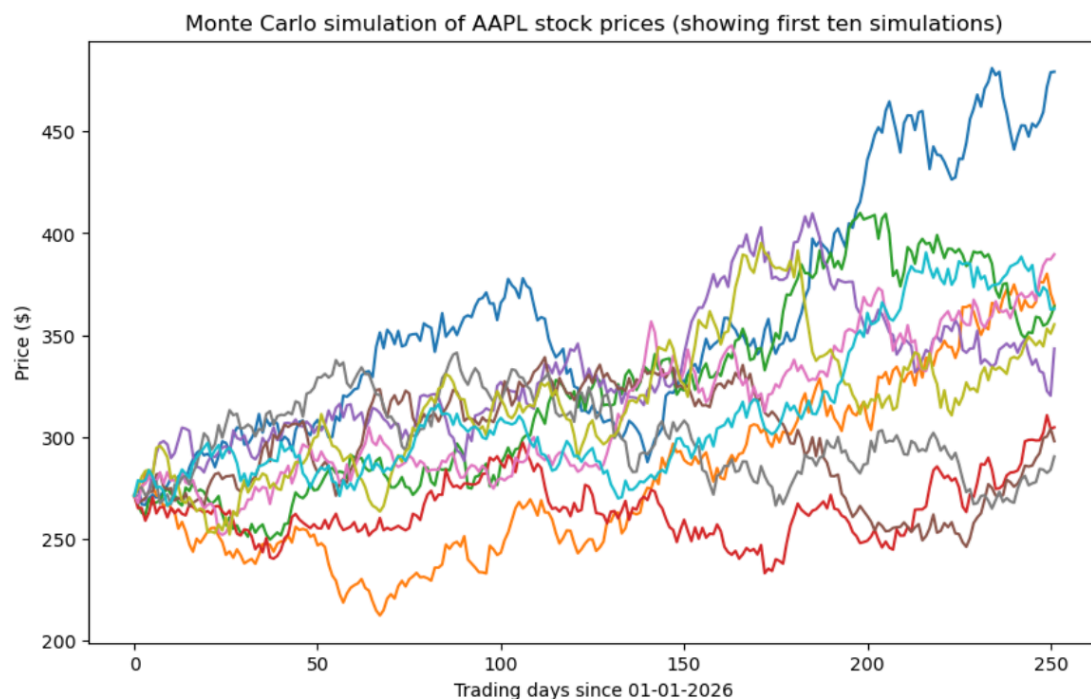
Random shocks (the instantiations of Z) were simulated using the normal distribution function from numpy.

4 - Results

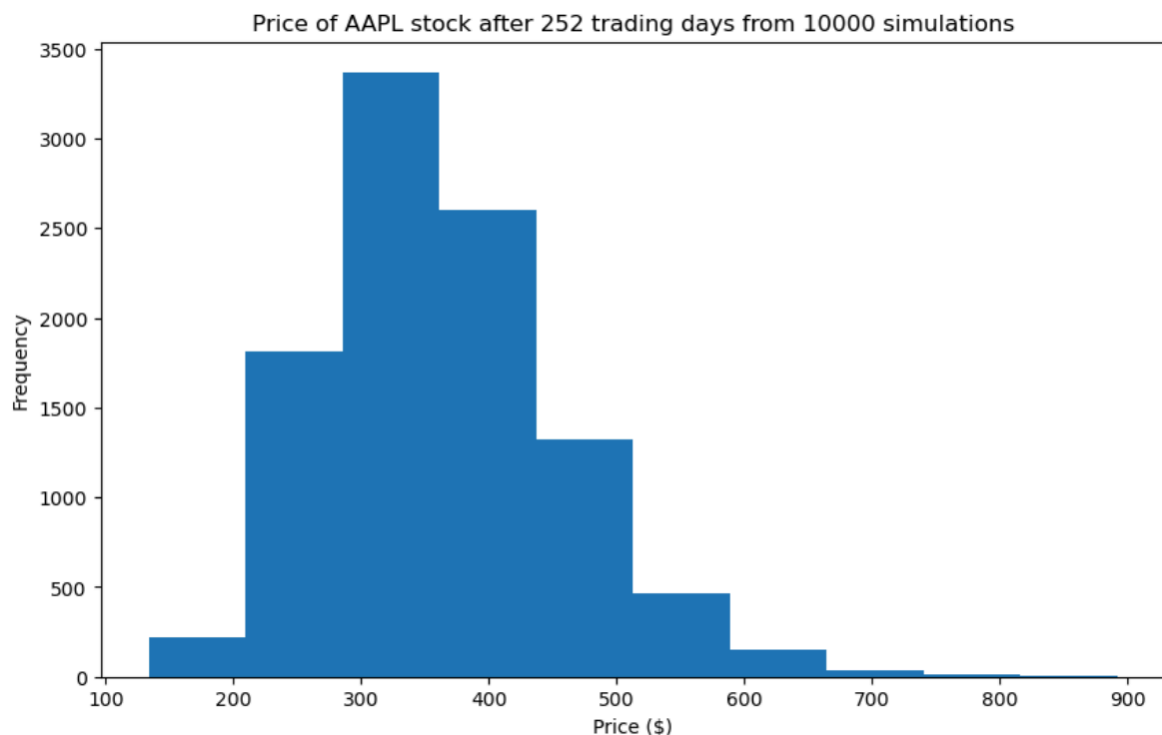
In order to obtain some of the following results, it was necessary to condense the main programme into a new function form which outputs just the list of terminal prices for the individual simulations. The document containing with code is available alongside that with the main code.

4.1 - General results

Below are two typical examples of the graphics generated by the programme.



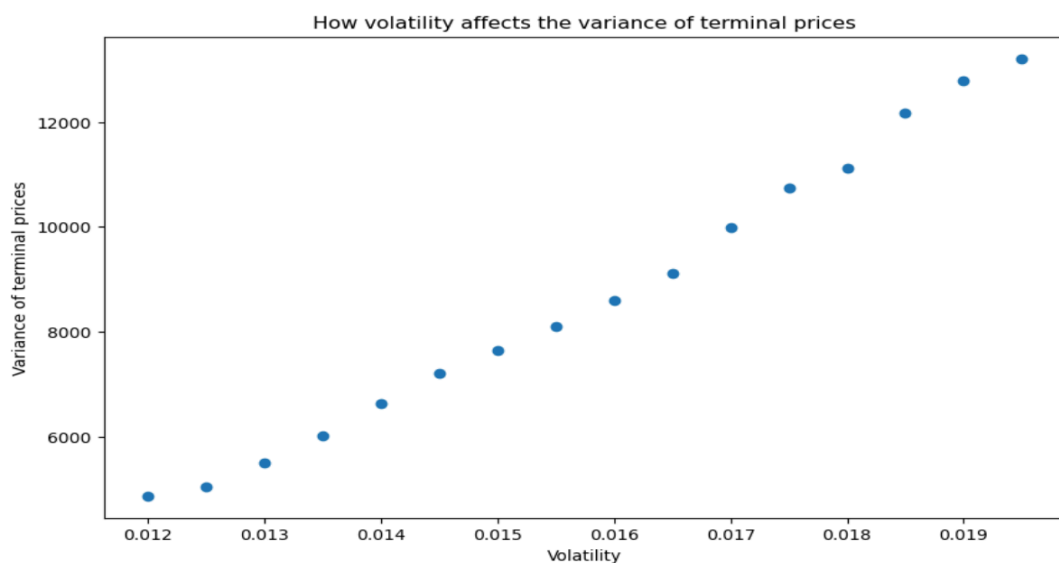
4.1.1 - Typical plot of first ten individual simulations (volatility = 0.01603, number of simulations = 10,000)



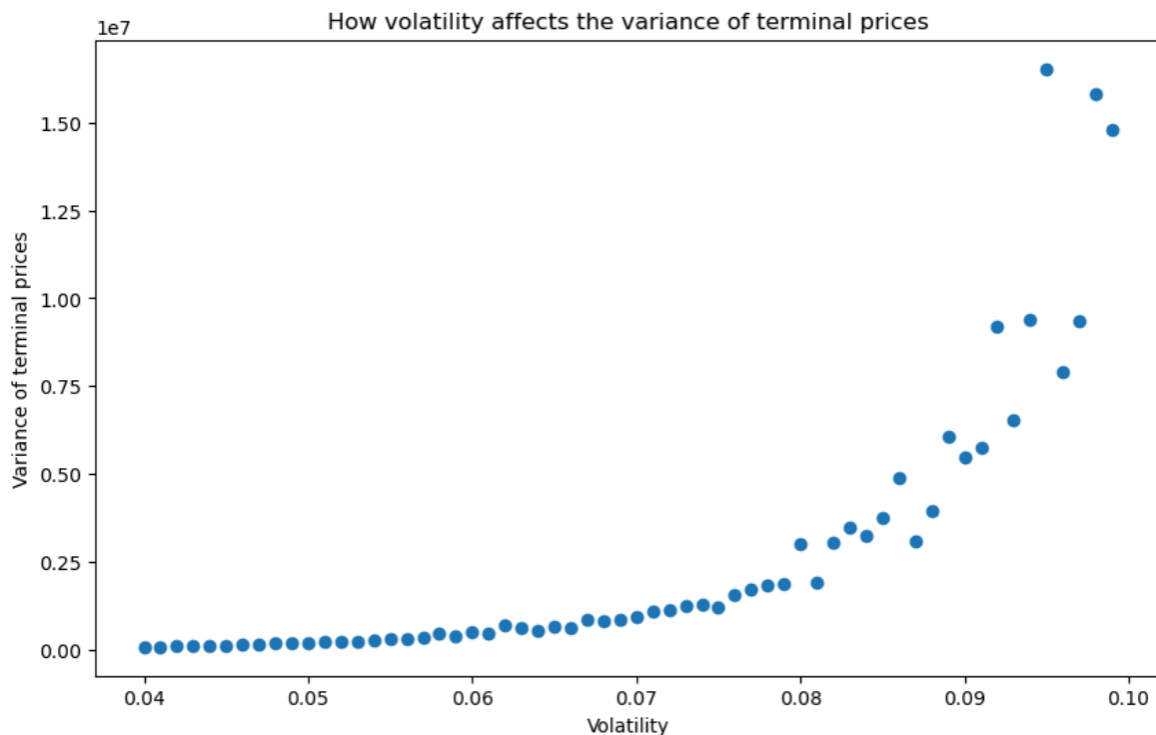
4.1.2 - Typical histogram (same simulation as 4.1.1)

Positive skew is clearly visible in 4.1.2. This is to be expected because any price is bounded by 0 to the negative but can, in principle, be arbitrarily high to the positive. An individual simulation trending maximally downwards can only ever get close to 0, but one trending maximally upwards can get as high as the length of the simulation allows.

4.2 - Volatility



4.2.1 - Effect of volatility in the local area around actual volatility as estimated from historical data



4.2.2 - Same plot as 4.2.1 but with a wider range of volatilities

As can be seen in 4.2.1, the impact of volatility on the spread of simulated terminal prices is pronounced. This is because the volatility is mathematically important in the GBM model, having an amplifying effect on the random shocks, which causes more divergent paths.

5 - Assumptions and Limitations

Firstly, an important shortcoming of the GBM model is that it assumes random shocks are normally distributed. In reality, asset returns have fat tails, meaning markets are subject to unusual but highly impactful 'black swan' events far more often than GBM predicts.

Secondly, the model is depends on extrapolating market behaviour from the past as if that is the general rule. For example, expected return and volatility are treated as constant rather than variable. This is highly unrealistic, meaning the model is likely only valid for short times after its input period.